



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

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ASSIGNMENT No : 7

TITLE :Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Problem Statement

Implement a Java program to demonstrate inheritance and interfaces using suitable real-world examples.

Learning Objectives

To implement inheritance and interfaces for achieving code reusability and abstraction. -----

Theory

Inheritance allows one class to acquire the properties and methods of another class, promoting code reuse and reducing redundancy. Java supports single inheritance through the extends keyword. Interfaces define a contract that implementing classes must follow, enabling abstraction and multiple inheritance of type. Together, inheritance and interfaces support polymorphism and modular software development. They are fundamental concepts of object-oriented programming.

Outcome

The program successfully implemented inheritance to reuse code between a parent and child class, and interfaces to enforce a contract for method implementation. Objects of derived classes correctly inherited and overrode parent behavior, and interface methods were implemented as expected.

Conclusion

Inheritance reduces code duplication and models real-world "is-a" relationships, while interfaces provide abstraction and support multiple inheritance of type. Together, they form the backbone of polymorphism and are essential for scalable, reusable, object-oriented design.

Exercise

1. Create a **Shape application** where Circle and Rectangle inherit from Shape and calculate area.

```
1  abstract class Shape {
2      |    abstract double area();
3      |
4  }
5  class Circle extends Shape {
6      |    double radius;
7      |    Circle(double radius) { this.radius = radius; }
8      |    double area() { return Math.PI * radius * radius; }
9      |
10     }
11     class Rectangle extends Shape {
12         |    double length, width;
13         |    Rectangle(double length, double width) { this.length = length; this.width = width; }
14         |    double area() { return length * width; }
15         |
16     }
17     public class ShapeDemo {
18         |    Run | Debug
19         |    public static void main(String[] args) {
20             |    Shape c = new Circle(radius: 5);
21             |    Shape r = new Rectangle(length: 4, width: 6);
22             |    System.out.println("Circle Area: " + c.area());
23             |    System.out.println("Rectangle Area: " + r.area());
24         |
25     }
```

```
Circle Area: 78.53981633974483
Rectangle Area: 24.0
```

2. Develop a **E-Commerce Product System** where electronic, clothing, and grocery products inherit common properties and implement product interfaces.

```
1  interface Product {
2      void display();
3  }
4  class Electronic implements Product {
5      String name = "Laptop";
6      public void display() { System.out.println("Electronic Product: " + name); }
7  }
8  class Clothing implements Product {
9      String name = "Shirt";
10     public void display() { System.out.println("Clothing Product: " + name); }
11 }
12 class Grocery implements Product {
13     String name = "Rice";
14     public void display() { System.out.println("Grocery Product: " + name); }
15 }
16 public class ECommerceDemo {
17     Run | Debug
18     public static void main(String[] args) {
19         Product[] products = { new Electronic(), new Clothing(), new Grocery() };
20         for (Product p : products) {
21             p.display();
22         }
23     }
24 }
```

```
Electronic Product: Laptop
Clothing Product: Shirt
Grocery Product: Rice
```
